

Fig. 5: System configuration

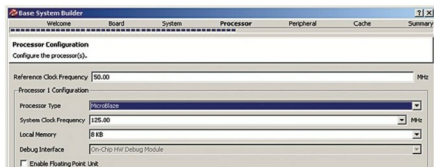


Fig. 6: Processor configuration

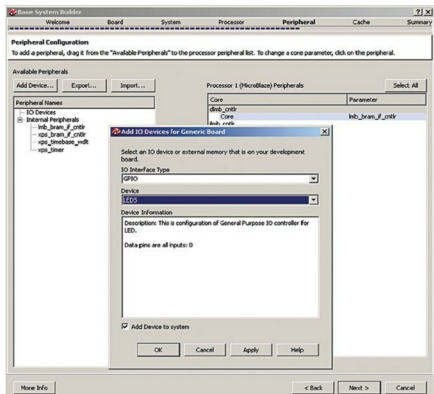


Fig. 7: Peripheral configuration

you want to build a single-processor system or a dual-processor system. Select Single-Processor System and click Next (Fig. 5)

8. The Processor Configuration page appears. Select 50.00MHz as the reference clock frequency. The Virstex-5 FPGAs do not have any hard processor, so Base System Builder will set up a MicroBlaze soft processor. Set the system clock frequency as 125.00MHz and local memory as 8KB, and click Next, as shown in Fig. 6

9. The Peripheral Configuration page appears through which we select the peripherals to put in the design. The peripherals will be connected to the MicroBlaze processor via the PLB, and they allow you to control and access features of the FPGA and external hardware. Click the Add Device button, select the IO Interface Type as GPIO and the device as LEDs, as shown in Fig. 7. Click OK

10. Select the GPIO Data Width as 8, as shown in Fig. 8, and click Next

11. Click Next on the Cache Configuration window

12. The Summary page (refer Fig. 9) gives you a summary of the design created by Base System Builder, showing the PLB memory map, peripherals and the files that will be generated

13. Since we are designing using a custom board, the window shown in Fig. 10 appears, indicating that the user needs to update the user constraints file and set the correct position of the FPGA in the JTAG chain. Click OK

14. The System Assembly View window appears, as shown in Fig. 11. The different regions in the window are explained below for understanding

15. The area marked as (A) contains the project information and has two tabs, namely, Project and IP Catalogue. The Project tab lists the project files like the Microprocessor Hardware Specification (MHS) file, User Constraints File (UCF), Bitgen Options File, iMPACT command file, implementation options file and the ELF file, and the project settings, including the device number, simulation model and netlist type

The MHS file defines the hardware